

Shubham Hadawle

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EDUCATION

Master of Science in Computer Science

Northeastern University, Boston

Coursework: Algorithms, Database Management Systems, Programming Design Paradigms, NLP, MLOps

Jan 2026 – Dec 2027

Bachelor of Engineering in Artificial Intelligence and Data Science

University of Mumbai, Mumbai

Coursework: Data Structures, Machine Learning, Deep Learning, Object Oriented Programming - Java

Jul 2020 – Jun 2024

PROFESSIONAL EXPERIENCE

Motilal Oswal Financial Services

Software Engineering Intern

Sep 2025 – Nov 2025

- Boosted the **retrieval quality** of a RAG system by **14%** by designing a hierarchical clustering pipeline for text embeddings (SBERT).
- Automated the processing of financial research PDFs via **LLM (Sonnet 4.5)** on **AWS Bedrock**, generating context-preserving clusters with 82% chunk coverage across a 50k+ document corpus.
- Integrated OCR and LandingAI's Agentic workflow (DPT-2) for PDF parsing to enhance **RAG accuracy** by 1.3 times.

Axis Bank

Software Engineering Intern

May 2025 – Aug 2025

- Devised a Microsoft **GraphRAG** system by integrating knowledge graphs with LLMs for scalable, multi-hop information retrieval.
- Accelerated large-scale financial data retrieval by **8 mins** by deploying a Text-to-Query **microservice** to convert natural language into executable Cypher queries.
- Reduced manual **SQL** querying overhead by **1/2** on customer datasets by leveraging **Neo4j** and structured schema graphs.

Bhabha Atomic Research Centre (BARC)

Software Engineering Intern

Nov 2023 – Jan 2024

- Designed Deep **Neural Networks** (LSTMs, Autoencoders, VAEs and Neural ODEs) to solve Point Kinetic Equations that simulated the behavior of a Uranium-235 Nuclear Reactor.
- Saved ~6 hours** of manual effort weekly by streamlining mathematical calculations via Python and ML libraries like **NumPy**, **Pandas**, **Matplotlib**, **Seaborn**, **Pyplot**, **Scikit learn**, **TensorFlow**, **Keras** and **PyTorch** (torchdyn/torchdiffeq).
- Optimized cross-domain adaptation using Adam optimizer to refine the loss function, reducing the loss from **1.19520 to 0.00062**.

Stocc Guru (FinTech startup)

Software Engineering Intern

Jul 2021 – Sep 2021

- Drove active engagement for **2,000+ live users** by developing a **full-stack Java Android** application, promoting stock market investments and financial literacy through real-time updates.
- Collaborated with cross-functional teams to implement **CI/CD** pipelines via **GitHub Actions**, accelerating release cycles and reducing post-deployment bugs by **11%**.
- Architected backend services around SOLID design principles and optimized JSON serialization, reducing **API latency** to **25 ms**.

PROJECTS AND PAPERS

Platformer Game: Where is my Duck? (1000+ Users on Android)

([App Link](#))

- Engineered and deployed a cross-platform 2D application using GDevelop5 for Android, Windows, Linux, and macOS, successfully acquiring **1,000+ downloads** on the Google Play Store and maintaining a **4.7/5 stars** average user rating.
- Monetized the Android app by integrating Google AdMob, conducting rigorous **A/B testing** on ad frequency and strategic placement of interstitial and banner ads to achieve a **15% increase in CTR** (click-through rate).

Generating Question Paper from Handwritten Notes using T5 Transformer

- Automated question paper generation from handwritten or textual notes by integrating **OCR** and **Google Cloud Vision API** from GCP with a fine-tuned **T5 Transformer**, speeding-up assessment creation by 60%.
- Built a Python system, hosted on **Amazon Elastic Beanstalk**, to enable generation of MCQ/short-answer/True-False questions, allowing educators to upload scanned notes and download formatted question papers (PDF/Word) in **< 2 minutes**.

TECHNICAL SKILLS

- Languages & Frameworks:** Python, Java, C, C++, SQL, JavaScript, Go, React, Node.js, Flask,
- Databases & Big Data:** MySQL, MongoDB, PostgreSQL, Firebase, Snowflake, Apache Spark, ETL
- Full-stack:** HTML, CSS, APIs, REST APIs, Postman, XML, Android Studio, Backend, Git, GitHub
- AI/ML:** PyTorch, TensorFlow, Keras, Scikit-Learn (sklearn), NumPy, Pandas, NLTK, LangChain, LLMs, GenAI, RAGs
- Cloud & DevOps:** AWS, Google Cloud Platform (GCP), Microsoft Azure, Docker, Kubernetes, Terraform, CI/CD
- Other Skills:** SDLC, Testing, Debugging, Distributed Systems, Microservices, Agile, Scrum, Linux, Unix, AI tools

RESEARCH PAPERS

- Text Summarization of Regional Languages using Self-Attention Mechanism – ([Publication Link](#))
- Anomaly Detection in Multi-variate Time Series Data using Facebook Prophet – ([Publication Link](#))